



Titration – The Secret Code of Concentration

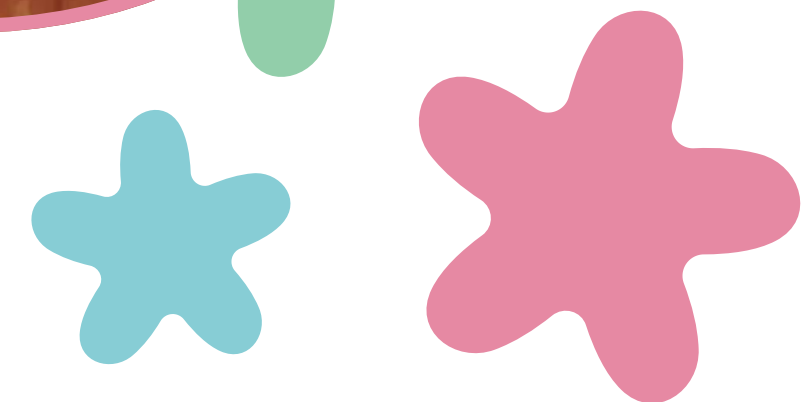


The Sour Lemonade Challenge!

Is this Nimbu Pani too sour or just right?

At the mela stall, every glass looks the same — but they all taste different! Some are too sour, some too sweet, and some just perfect. How do we find out what's inside?

Today, we become taste detectives! 🍋 We will learn how scientists figure out exactly how strong or weak a drink really is — drop by drop!





What is Concentration?

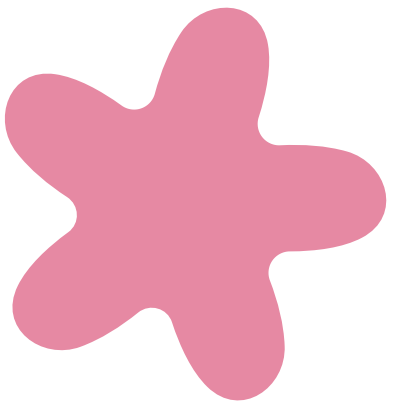
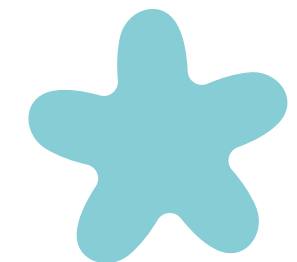
Concentration means how strong or weak a drink is! Think about squeezing lots of lemons into a tiny glass — that makes it super strong and sour. Add more water and it becomes weak and mild!

Strong lemon juice has lots of lemon packed in. Watery lemonade has very little. Sweet Rasna syrup is very concentrated before you add water. More stuff dissolved = stronger concentration!

Meet the Titration Tools

Every science detective needs the right tools! In titration, we use special equipment to find out how strong or weak a liquid is — drop by drop!

-  Burette — holds the liquid and releases drops
-  Dropper — adds liquid carefully
-  Flask — holds the mystery liquid
-  Indicator — changes color to give us the answer!





The Color-Changing Helper: Indicator

An indicator is a magic helper that changes color when mixed with acids or bases! Watch as just one drop transforms the whole flask from yellow to bright pink — that's the secret signal we've been waiting for!

Try it at home! Turmeric turns bright yellow in acid and greenish-blue in a base. Red cabbage juice turns purple in base and pink in acid. Nature's own color-changing indicators are all around us!



Virtual Kitchen Science Experiment

Let's mix lemon and baking soda with turmeric! Watch what happens when you add drops one by one into the flask. The turmeric changes from yellow to greenish-blue — that's the magic of titration happening right in your kitchen!

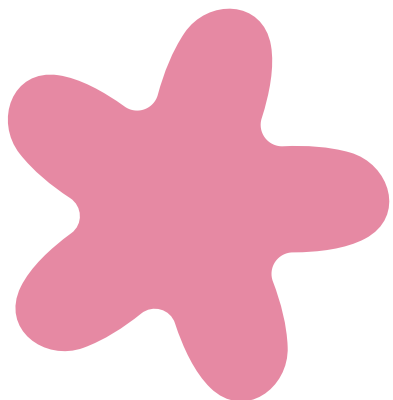
Step 1: Pour turmeric water into the flask.

Step 2: Add lemon drops slowly — count each one!

Step 3: Watch the color shift from yellow to green.

Step 4: Add baking soda — see it change back!

"Add drops, watch color change!"



Master Drop Counter Challenge!

Can you find the magic drop? Count carefully!

● **Flask A:** Add drops of lemon juice into baking soda + turmeric.

How many drops to change color from yellow to green?

☐ 3 drops ☐ 5 drops

● **Flask B:** Add drops of red cabbage juice into soap water.

How many drops to see the color switch to purple?

☐ 2 drops ☐ 4 drops

★ **Tip:** Watch carefully — the endpoint is just ONE magic drop!





The Magic Endpoint — The Color Switch!

The endpoint is the magic moment when the indicator changes color — and it happens with just **ONE** final drop! This is the secret code that tells us we have found the exact concentration. The color switch means: perfect balance achieved!

Think of it like a seesaw — too much acid on one side, too much base on the other. At the endpoint, the seesaw is perfectly balanced! The flask turns from yellow to bright pink, and that's our "just right" signal. Drop by drop, we crack the secret code!





Why Does Titration Matter?

-  Check if milk or honey is pure and safe to drink!
-  Help farmers know if their soil is healthy for crops.
-  Test if river or pond water is safe for us and animals.

Titration is used every day by scientists, farmers, and doctors in India to keep us safe and healthy!



Thank You!

Try safe kitchen
science at home!

